Unsteady SIMPLE Solver — Verification, SIMPLEC/SIMPLER and Central Differencing

Overview

You will verify an unsteady finite volume solver against exact solutions, then extend it in two directions: two new pressure–velocity coupling algorithms (SIMPLEC and SIMPLER), and a second-order convection scheme (central differencing). The backward-facing step is the common test case for both extensions.

Code provided. unsteady_SIMPLE_FVM.m solves 2D unsteady incompressible laminar flow on a staggered grid with SIMPLE, first-order upwind convection and implicit Euler time stepping. Section 1 (CASE SETUP) is the only part that changes between problems:

Setting	Options
caseName	'channel', 'couette', 'stokes1', 'stokes2'
xBC	'inlet-outlet' (velocity inlet uInlet(y,t), zero-gradient outlet) or 'periodic'
southBC, northBC	'wall' (moving tangentially with uSouth(t) or uNorth(t)) or 'slip' (zero shear)
dpdx	imposed mean pressure gradient (periodic cases)
dt, tEnd, tOut, stopWhenSteady	time step, end time, storage times, stop once steady

Keep the structure of the code and add new options (algorithm = 'SIMPLEC', algorithm = 'SIMPLER', and scheme = 'CDS') rather than writing separate programs, so that every comparison uses the same grid, boundary conditions and post-processing. Work in MATLAB R2016b or newer, or GNU Octave. Every number in your report must come from code you submit.

Part A — Verification with exact solutions (25 marks)

Run the three cases as supplied, then test how the error depends on the time step, the grid and the domain.

A1. Developing channel flow (8 marks). caseName = 'channel': $Re = \rho U_o H/\mu = 100$, fluid initially at rest, uniform inflow.

1. Plot the outlet profile at each stored time against the Poiseuille profile $u = 6U_o(y/H)(1 - y/H)$, and the centreline velocity along x.
2. Define the entrance length as the x where the centreline velocity reaches 99% of $1.5U_o$. Compare it with the correlation $L_e \approx 0.05\ Re\ H$.

3. Halve dx only, then dy only, and report the change in L_e and in the outlet error. Which direction controls each quantity? Relate this to the cell Péclet number $U_o\Delta x/\nu$ and to the false diffusion of upwinding, about $U_o\Delta x/2$.
4. Run with dt = 0.1, 0.25 and 1.0. Does the steady solution depend on dt? Does the time to reach it?

A2. Stokes first problem (8 marks). caseName = 'stokes1': plate impulsively started with speed U_o.

1. Compare solution with the exact solution.
2. Plot u/U_o against $\eta = y/(2\sqrt{(\nu t)})$ for all stored times and show that they collapse onto one curve.
3. Compute the wall shear stress from your solution and compare it with $\tau_w = -\mu U_o/\sqrt{(\pi\nu t)}$ on log–log axes.
4. Time-step study: run dt = 0.004, 0.002, 0.001 and 0.0005 with the grid fixed. Plot the maximum error at t = 0.25 s against dt and find the observed order. Then do the same for the grid, with dt small. Which error dominates at early times, and why?
5. Reduce H until the far-field boundary visibly changes the profile at t = 1 s. Compare that H with the 99% penetration depth $3.64\sqrt{(\nu t)}$.

A3. Stokes second problem (9 marks). caseName = 'stokes2': plate oscillating as $U_o\cos \omega t$.

1. Compare result with exact solution.
2. Plot the profiles at eight phases of the last period with the exact curves and the envelope ±e^{−ky}.
3. Show that the wall shear stress leads the plate velocity by 45° by plotting both over the last period.
4. Run 50, 100, 200 and 400 time steps per period. For each, measure the amplitude and the phase of u at $y = \delta$, from a fit over the last period, and compare them with the exact values e^{−1} and a lag of 1 rad. Explain the trend using the amplification factor of implicit Euler for a single Fourier mode.
5. How many cells per δ are needed for a 1% amplitude error at $y = \delta$?

Questions to answer in the report.

1. In the periodic cases v = 0 and p is uniform. Show from the discrete equations why, and state which terms of the Navier–Stokes equations these cases test and which they do not.
2. Why does the Stokes first problem need far fewer SIMPLE iterations per step at large t than at small t?
3. At the same dt, the Stokes second problem needs about 15 SIMPLE iterations per step. Which setting in the code controls this, and what would bring it down?

Part B — Backward-facing step set-up (10 marks)

Add a case caseName = 'bfs' using only Section 1 of the code. No change to the solver is needed: with xBC = 'inlet-outlet', an inlet profile that is zero below the step lip makes that part of the west boundary a no-slip wall, the step face.

Geometry and conditions (Armaly et al., 1983).

- Step height S = 4.9 mm, inlet channel height h = 5.2 mm, outlet height H = S + h, expansion ratio H/h = 1.94. Work with lengths scaled by S: S = 1, h = 17/16.
- Domain from the step (x = 0) to x = 40 S. Grid: 16 cells across the step and 17 across the inlet (Ny = 33), with $\Delta x = 2\Delta y$.
- Inlet: fully developed parabola of mean velocity U_o for S < y < H, and u = 0 for y < S.
- Walls at y = 0 and y = H; zero-gradient outlet (already in the code).
- Reynolds number as Armaly defined it, from the inlet hydraulic diameter: $Re = \rho U_o(2h)/\mu$, so set μ from Re.
- March in time from rest to steady state (stopWhenSteady = true).

B1. Write the case block and run Re = 100. Plot streamlines, shading the recirculating fluid.

B2. Write a function that returns the reattachment length x_1 (where u in the first cell row changes from negative to positive, by linear interpolation) and, when present, the separation and reattachment points x_4 and x_5 of the upper-wall bubble.

B3. Find a time step that reliably reaches steady state at Re = 400 and Re = 800. Report what happens with too large a time step, especially in the first steps after an impulsive start, and how you detected it.

Reference data for Parts C and D: Armaly's primary reattachment lengths, as tabulated against $Re_S = U_oS/\nu$ in the Feel++ laminar backward-facing-step benchmark, converted to $Re = U_o(2h)/\nu$.

Re	72	102	174	300	350	497	632
x_1/S (experiment)	2.48	3.05	4.36	6.73	7.97	9.55	11.47

Armaly et al. report three-dimensional flow in their channel above Re ≈ 400. For a 2D numerical comparison at Re = 800, Gartling (1990) gives $x_1/S = 12.2$ and an upper-wall bubble from x/S = 9.7 to 21.0, for an expansion ratio of 2.

Part C — SIMPLEC and SIMPLER (25 marks)

Add an option algorithm = 'SIMPLE' | 'SIMPLEC' | 'SIMPLER' to Section 1. Change only the pressure–velocity coupling; keep upwind convection and implicit Euler. Modify the code to demonstrate SIMPLEC and SIMPLER algorithms.

C1. SIMPLEC: Show derivation

C2. SIMPLER (10 marks): Show derivation

C3. Tests (7 marks).

1. Stokes second problem, 200 steps per period: tabulate the mean number of iterations per time step and the CPU time for the three algorithms at α_u = 0.5, 0.8 and 1.0. For SIMPLE use $\alpha_p = 0.3$. Confirm that the three solutions are identical to within the tolerance, and explain what the iteration counts show.
2. Backward-facing step at Re = 400 and Re = 800: march to steady state with each algorithm. Report iterations per step, total CPU time to steady state, and x_1, x_4 and x_5. Find the largest stable dt for each algorithm.

3. Plot the SIMPLE-iteration count per step against time for the three algorithms in one figure for the Re = 800 step.

Questions to answer in the report.

1. Do the three algorithms give different results? Answer for each time step of a transient run and for the final steady state, and justify the answer from the discrete equations.
2. Why does the advantage of SIMPLEC and SIMPLER grow as dt increases?
3. Which algorithm would you use for (a) a Stokes-type problem with small dt, (b) marching to a steady state with large dt? Give the evidence.
4. Why can the Stokes problems not distinguish the three algorithms, while the backward-facing step can? What does that tell you about choosing test cases for a coupling algorithm?

Part D — Second-order central differencing (25 marks)

Add an option scheme = 'UDS' | 'CDS' | 'CDS-DC' for the convective terms of both momentum equations. Use SIMPLEC from Part C.

UDS – Upwind Difference Scheme **(already available)**

D1. CDS – Central Difference Scheme: At a face that lies on a boundary (the v control volumes next to the inlet, for example) use the boundary value, not an interpolation. Run the step at Re = 100, 400 and 800 on the Part B grid. Map the cell Péclet numbers F/D in x and y, report where they exceed 2, and describe what the solution and the SIMPLE iterations do there: convergence, wiggles, negative coefficients.

D2. CDS-DC – Central Difference Scheme – Deferred Correction: Keep the upwind coefficients in the matrix, and add the difference between central and upwind face fluxes as an explicit source, evaluated with the latest iterate (Khosla and Rubin, 1974).

D3. Performance on the backward-facing step (10 marks).

1. Grid study at Re = 400 on two grids, 16 + 17 and 32 + 34 cells across the height, with $\Delta x = 2\Delta y$ (both keep the expansion ratio exact). Tabulate x_1/S for UDS and CDS-DC, and Richardson-extrapolate x_1 assuming each scheme's formal order. Compare the two extrapolated values. Optional: add a 64 + 68 grid and measure the observed orders.
2. Reynolds-number sweep Re = 100, 200, …, 800 on the middle grid with UDS and CDS-DC. Plot x_1/S against Re with Armaly's data and Gartling's 2D value, and plot x_4/S and x_5/S.
3. Report the cost of each scheme: iterations per step and CPU time to steady state.

Questions to answer in the report.

1. Why does upwind differencing shorten the recirculation zone, increasingly so as Re rises? Estimate the ratio of numerical to physical viscosity, $U_0\Delta x/(2\nu)$, on your grids.
2. Why does direct CDS lose boundedness when the cell Péclet number exceeds 2? Relate this to the signs of the coefficients and to the Scarborough criterion.
3. Does deferred correction change the converged solution, or only the convergence path? How does this parallel your answer on SIMPLE, SIMPLEC and SIMPLER in Part C?

4. Up to what Re do your results follow Armaly's data? Separate the numerical part of the discrepancy (from D3.1) from the modelling part (2D versus 3D flow, inlet profile imposed at the step).

Deliverables and marking

Submit one zip file containing your modified Matlab package (all options in one file), any helper functions and driver scripts, and a report of at most 15 pages. The code must run as submitted.

Part	What is assessed	Marks
A	Verification plots, order-of-accuracy results, answers to questions 1–3	25
B	bfs case block, reattachment function, time-step findings	15
C	SIMPLEC and SIMPLER code, iteration and cost tables, answers to questions 1–3	30
D	CDS and CDS-DC code, Péclet map, grid study, x_1/S against Armaly, answers to questions 1–4	30

Marks for answers go to the reasoning: a correct number with no explanation earns little. The fine grid of D3.1 (32 + 34 cells across the height, about 43 000 cells) can take a long time to march to steady state; start it early.